Practice workbook · Simultaneous equations with powers and logs

Turning a pair of index and log equations into ordinary algebraic equations in x and y and solving them together.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-12

1. Question 9 · part whole · 8 marks

2023 JAN · Paper 01

Worked-example retry · Attempt part whole only.

9 Giving each value in your solution to 2 decimal places, solve the simultaneous equations

$$e^{2y} - x + 2 = 0$$

$$\ln(x + 3) - 2y - 1 = 0$$

(8)

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1. Question 9 · part whole · 8 marks

2023 JAN · Paper 01

Worked-example retry · Attempt part whole only.

Question 9 continued

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(Total for Question 9 is 8 marks)

1. Question 9 · part whole · 8 marks · Working and answer

Working and answer

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2. Question 10 · part b · 9 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

10 (a) Show that $\frac{9^{3y}}{243} = 3^{(6y-5)}$

(4)

(b) Solve the simultaneous equations

$$\frac{9^{3y}}{243} = 27^{(x-2)}$$

$$\log_{10}\sqrt{6xy} = \log_4 2$$

(9)

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2. Question 10 · part b · 9 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

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2. Question 10 · part b · 9 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

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2. Question 10 · part b · 9 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

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(Total for Question 10 is 13 marks)

2. Question 10 · part b · 9 marks · Working and answer

Working and answer

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3. Question 8 · part whole · 8 marks

2021 JUN · Paper 02

Further practice · Attempt part whole only.

8 Use an algebraic method to solve the simultaneous equations

$$\log_4 a + 3\log_8 b = \frac{5}{2}$$

$$2^a = \frac{16^4}{4^{b^2}}$$

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3. Question 8 · part whole · 8 marks

2021 JUN · Paper 02

Further practice · Attempt part whole only.

Question 8 continued

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(Total for Question 8 is 8 marks)

3. Question 8 · part whole · 8 marks · Working and answer

Working and answer

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4. Question 8 · part whole · 8 marks

2025 NOV · Paper 02

Further practice · Attempt part whole only.

8 Solve, using an algebraic method, the simultaneous equations

$$\log_{27} x^3 - \log_3 y = 2$$
$$\log_6 (x + 3y) = 3$$

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4. Question 8 · part whole · 8 marks

2025 NOV · Paper 02

Further practice · Attempt part whole only.

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Question 8 continued

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(Total for Question 8 is 8 marks)

4. Question 8 · part whole · 8 marks · Working and answer

Working and answer

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5. Question 9 · part whole · 7 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part whole only.

9 Showing your working clearly, use algebra to solve the equations

$$\frac{16^x}{8^y} = \frac{1}{4}$$

$$4^x 2^y = 16$$

(7)

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5. Question 9 · part whole · 7 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

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(Total for Question 9 is 7 marks)

5. Question 9 · part whole · 7 marks · Working and answer

Working and answer

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6. Question 11 · part whole · 8 marks

2023 NOV · Paper 01

Further practice · Attempt part whole only.

11 Solve the simultaneous equations

$$2\log_4 x = \log_3 3y^2$$

$$\log_2 x^3 + 8\log_9 y = 13$$

Show your working clearly.

(8)

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6. Question 11 · part whole · 8 marks

2023 NOV · Paper 01

Further practice · Attempt part whole only.

Question 11 continued

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6. Question 11 · part whole · 8 marks

2023 NOV · Paper 01

Further practice · Attempt part whole only.

Question 11 continued

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(Total for Question 11 is 8 marks)

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6. Question 11 · part whole · 8 marks · Working and answer

Working and answer

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